Sutram SRM — Sentinel-2 10 m → 2.5 m

Deep-learning super-resolution with per-pixel trust · Team Sutram · Smart India Hackathon 2026

0.917	**1st**	**0.00 m**	**30 ms**
FIELD-BOUNDARY F1	ON EVERY METRIC	GEO DRIFT	PER TILE

THE PROBLEM

Sentinel-2 images all of India every 5 days, free, at 10 m — enough to see a field, not its boundary. Commercial sub-metre imagery costs thousands of rupees per scene.

Generative super-resolution offers a third path and introduces a new risk: **a model that sharpens can also invent.** That risk, not image quality, is why SR is not already operational in most agencies.

> **What we actually solve**
> Not "make images sharper" — open models already do that. The unsolved problem is making the result **trustworthy enough to act on**.

PIPELINE

```
S2 L2A (B04 B03 B02 B08 @10m + SCL)
  |
[1] SCL cloud mask -> scale -> tile 128/32
  |
[2] Branches (one interface)
     bicubic | SEN2SR | LDSR-S2 | OURS
  |
[3] Trust layer
     LR-consistency | spectral angle
     branch spread  | sampling sigma
       -> confidence [0,1]
  |
[4] Wald protocol | consistency | calibration
  |
[5] 2.5m COG: 4 bands + sigma + confidence
  |
[6] NDVI | buildings | change detection
```

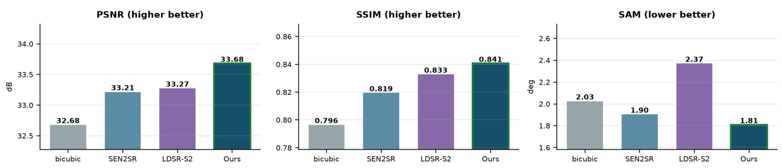
TRAINING DATA

Site	Location	Patches
KUDALIAR	**Telangana, INDIA**	**4,000**
ANJI	China	2,312
MAD-AMBO	Madagascar	1,442
SUDOUE-4	France	935
ESTUAMAR	Spain	911
Train total		**8,640**

SEN2VENuS — real **same-day** Sentinel-2/VENuS pairs. India capped at 4,000 so it cannot dominate. 50 epochs, 41 min on Apple Silicon.

> **Scale subtlety, stated openly**
> SEN2VENuS is natively ×2 (10→5 m). We build a ×4 problem by degrading the *input* to 20 m through the Sentinel-2 PSF — not bicubic — while keeping the **real** 5 m target.

RESULTS — WALD PROTOCOL, CLEAN TILE ~1200 KM FROM TRAINING

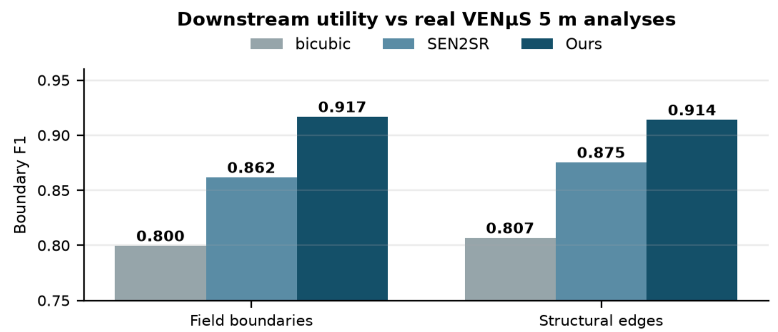


Branch	PSNR	SSIM	SAM°	ERGAS	Time
bicubic	32.68	0.7965	2.025	3.735	<1 ms
SEN2SR (ESA, RSE 2025)	33.21	0.8194	1.904	3.500	40 ms
LDSR-S2 (JSTARS 2025)	33.27	0.8328	2.370	3.506	105.6 s
OURS	**33.68**	**0.8406**	**1.808**	**3.331**	**30 ms**

RESULTS — SAME-SENSOR S2→S2, 400 HELD-OUT PATCHES

Branch	PSNR	SSIM	SAM°	RMSE
bicubic	35.580	0.8702	2.917	0.01710
SEN2SR	36.395	0.8929	2.704	0.01555
OURS	**37.182**	**0.9101**	**2.462**	**0.01421**

THE CLAIM THAT MATTERS — DOES THE ANALYSIS IMPROVE?



> **Not "sharper" — "better analysis"**
> Field-boundary delineation **0.800** → **0.917** (+14.7%), structural edges **0.807** → **0.914** (+13.3%), measured against analyses of **real VENuS 5 m imagery** — no degradation operator involved.

FEATURE COMPARISON

Capability	SEN2SR	LDSR	Ours
Per-pixel uncertainty	no	sigma	**4-signal**
Calibration validated	**no**	**no**	**YES**
Spectral angle optimised	no	no	**yes**
Interactive speed	yes	no	**yes**
Cloud masking built in	no	no	**yes**
Footprint verified	no	no	**0.00 m**
Downstream task eval	**no**	**no**	**YES**
Indian training data	none	none	**4,000**

APPLICATIONS

- **Agriculture** — plot-level crop health, sowing dates, PM-FBY insurance verification. Field boundaries **+14.7%**; NDVI preserved to **0.4%**.
- **Urban planning** — sprawl monitoring, unauthorised construction, road updating. Structural edges **+13.3%**.
- **Disaster assessment** — flood extent, crop loss, landslides. Two-date change signal preserved to **0.4%** while boundaries sharpen.

WHAT WE BUILT, WHAT WE STOOD ON

Component	Origin
SPAN backbone	ESA's, not ours
Initial weights	SEN2SRLite
Trained weights	**OURS**
Training regime	**OURS**
Loss function	**OURS**
Trust layer	**OURS**

We did not invent an architecture. We asked a different question — not how to sharpen better, but how to make sharpening trustworthy. **The result beats the ESA baseline whose backbone we borrowed, by training and objective alone.**

CAVEATS WE RAISE OURSELVES

> **Tested on our own degradation operator.**
> The Wald benchmark uses the same S2 PSF our training pairs were built from; SEN2SR was not. The downstream result avoids this.
> **Telangana scenes are training ground** (tile 44QKE) — demonstration only, never a quantitative claim.
> **Calibration is weak** (r = −0.23): top decile has 3.4× lower error, but it is not a fine-grained predictor.
> **ERGAS worse on one instrument.** Unresolved; not claimed.

> **We found our own biggest error**
> A Hann-window bug zeroed the first row/column of every tiled output — 4.5 dB, making bicubic appear to beat every learned model. We published that for two weeks. Caught by refusing a result that made no mechanical sense; every number here was re-measured after.